

BHARAT PULI

Ramachandrapuram, India, +91 7997257754, pulibharat2007@gmail.com, [LinkedIn](#), [GitHub](#), [Codolio](#)

PROFESSIONAL SUMMARY

AI & Machine Learning undergraduate with research experience at **IIT Ropar**, skilled in **software engineering, system design, and full-stack ML deployment**. Proficient in **Python, Java, and C** with strong foundations in **Data Structures, Algorithms, and Object-Oriented Programming**. Experienced in building and shipping production-grade software using **FastAPI, Docker, PyTorch, and WebSockets**, with a track record of collaborative, real-world engineering across research and industry settings.

EXPERIENCE

Summer Research Intern — VLED Lab, IIT Ropar

May 2026 – Present

- Engineered a real-time webcam monitoring pipeline integrating face detection, phone/object detection, and eye-gaze tracking for an online learning platform, improving focus-violation detection accuracy by **15%**.
- Collaborated with senior researchers to design, build, and ship open-source ML software, accelerating research-to-delivery timelines across multiple projects.

AI and ML Intern — Elevate Labs

May 2025 – Jun 2025

- Designed, implemented, and evaluated supervised learning models in Python applying feature engineering, cross-validation, and precision-recall analysis across multiple production datasets.
- Recognized as **Best Performer** among 500+ global participants for technical innovation and software engineering excellence.

SKILLS

Languages: Python, Java, C, SQL

CS Fundamentals: Data Structures & Algorithms (250+ problems), OOP, Software Design, Testing & Debugging, Version Control (Git)

Frameworks & Tools: PyTorch, Scikit-learn, FastAPI, Flask, Docker, OpenCV, MediaPipe, YOLOv8, NumPy, Pandas, Streamlit, Hugging Face, Linux, Kaggle

ML & AI: Deep Learning, Reinforcement Learning, CNNs, RNNs, NLP, Computer Vision, Feature Engineering

EDUCATION

Vishnu Institute of Technology, Bhimavaram — B.Tech, AI & ML

CGPA: 9.1/10 | 2028

Coursework: Data Structures and Algorithms, OOP, Probability and Statistics, Linear Algebra, Machine Learning

Vignan Junior College — Intermediate (MPC)

95.7% | 2024

PROJECTS

CampusEye — Real-Time AI Surveillance System | [GitHub](#)

FastAPI, YOLOv8, WebSockets, SORT, Docker

- Built a real-time multi-object detection and tracking system using YOLOv8 + SORT, served via async FastAPI/WebSockets with live MJPEG streaming and Cloudflare Tunnel for remote access.
- Reduced identity-tracking errors across concurrent video streams; delivered a fully Dockerized, production-ready surveillance platform. [\[Demo\]](#)

Neural Style Transfer Web Application | [GitHub](#)

PyTorch, AdaIN, VGG-19, Flask, Docker

- Developed an artistic style transfer system using AdaIN + VGG-19 feature extraction, containerized with Docker and deployed on Hugging Face Spaces for zero-downtime public access.
- Achieved **90% user satisfaction** across **100+ validations**, with consistent content-structure preservation across diverse style inputs. [\[Live\]](#)

Flappy Bird Deep Reinforcement Learning Agent | [GitHub](#)

PyTorch, Deep Q-Network, Gymnasium

- Trained a DQN agent using experience replay and target-network sync; resolved 6 critical bugs (observation mismatch, sync instability, truncation) to stabilize convergence over 59,000 episodes.
- Achieved a best score of **131 pipes** — a **26× improvement** over the untrained baseline. [\[Demo\]](#)

ACHIEVEMENTS

Aspire Leaders Program (Aspire Institute): Selected from 45,228 global applicants; completed intensive international leadership development program.

Open Source Contributor (OSCG): Reduced ML software processing time by **20%** through algorithm optimization in collaboration with researchers.

GMI Mentorship Program: Selected for 2025 cohort focused on technology careers and professional growth.

CERTIFICATIONS

- AWS AI & ML Scholars Challenge** — Udacity, 2026
- Data Analysis with Python** — IBM / Coursera
- AI Foundations** — IBM SkillsBuild
- Data Analytics Job Simulation** — Deloitte (Forage), 2025